

Test Report - VxMark Drop Test, Bench Handling

Pius Wong , created Apr 7, 2026 updated May 28, 2026

Summary

VxMark v4.1 was found to resist failures and damage from expected drops during bench handling. However, recommendations were made for minimizing the effects of the headphones storage compartment door opening due to a drop.

Applicable VVSG Requirements

Primary VVSG requirement tested:

- 2.7-D – Ability to support maintenance and repair physical environment conditions – non-operating
 - TA2.7-D 1: MIL-STD-810H, Method 516.8, Procedure VI – Bench Handling

Supporting VVSG requirement tested:

- 2.7-E – Ability to support transport and storage physical environment conditions – non-operating

Devices Under Test

- VxMark, v4.1, MK-14-018

Purpose

VotingWorks is analyzing how well VxMark withstands drop tests required by VVSG, defined in MIL-STD-810H, Method 516.8: Bench Handling (Procedure VI). This is part of general data on how it withstands drops, impacts, and mechanical shocks from use, maintenance, and transportation.

This test also includes drop-testing some removable peripherals of the headphones and accessible controller. This helps meet the requirement to “configure the item as it would be for servicing,” listed in the MIL spec.

This test report is on the VxMark main control module, and not the COTS printer nor UPS used with the unit. Any concerns found here inform improvements to the VxMark design and user instructions. A passing test does not eliminate the need to test newer design iterations.

Background

VxMark units had not previously undergone standard drop tests. Informally, an internal accidental drop of a closed VxMark from a desk to the floor did not result in functional problems. However this large drop caused the touchscreen to scratch from contacting a socket head cap screw in the top tub. More information is needed about drop tests from bench handling.

It is recommended to perform other potentially destructive mechanical tests in combination with this drop test to provide more worst-case testing scenarios beyond VVSG.

Materials

1. Workbench and tools

- a. Hard tabletop
- b. Tools for opening unit as needed (metric hex wrenches, pliers, etc)
- c. Measurement tools for ensuring drops from 4 inches, and from a 45° angle.

2. Test unit, testing tools, and documentation tools

- a. VxMark
- b. Printout of normal function checklist (*VxMark Tests of Normal Function*), along with associated tools, including:
 - i. VxMark printer
 - ii. Appropriate paper within specifications for the system
 - 1. Bond 28-47lb / 105-177gsm
 - 2. 8.5" width; 11-14" length
 - iii. Metric hex wrenches, pliers, tools for testing fastener security
 - iv. light, camera
- c. Drop height and drop angle measurement tools

Procedure

The test procedure below is based on the bench handling drop test procedure in MIL-STD-810H cited in the VVSG2.0 requirements.

1. Check the unit for normal function before testing.

- a. Follow standard tests to confirm the test unit functions normally before any disruptive tests by following the normal function procedure. For the VxMark unit, normal functions include marking and printing ballots (when connected to a USB printer), giving visual and audio feedback, accepting touchscreen inputs, accepting PAT input, accepting tactile controller input, and securing internal components.
- b. Document tests of normal function. Record any pre-existing problems. Use the tests of normal function checklist template.
- c. For long-term tracking of units, record the life cycle history of the unit.

2. Execute drops on the closed and locked configuration.

- a. Refer to the standard procedure for drop tests on the closed configuration, described in MIL-STD-810H, Method 516.8: Bench Handling (Procedure VI). This is summarized in the next steps.
- b. Configure the test unit on the testing workbench or table in a closed and locked configuration (pictured below).



- c. Using one edge as a pivot, lift the opposite edge of the chassis until one of the following conditions occurs (whichever occurs first).
 - i. The lifted edge of the chassis has been raised 100 mm (4 in.) above the horizontal bench top.
 - ii. The chassis forms an angle of 45° with the horizontal bench top.
 - iii. The lifted edge of the chassis is just below the point of perfect balance.
- d. Drop the unit freely onto the horizontal bench top. Repeat using other practical edges of the same horizontal face as pivot points, for a total of four drops.
- e. Repeat the drops along each edge on each face, until it has been dropped for a total of at least four times on each face.
- f. Visually inspect the test item, and listen for loose parts, between drops and after all drops are done.
- g. Document the results.

3. Execute drops on any removable subsystem.

- a. Remove any subsystem from the test unit that is normally meant for users to remove.
- b. Follow the standard procedure for drop tests on each of the removable subsystems (listed above in Step 2). Inspect the subsystem for any changes, and document the results.

4. Check the unit for normal function after initial drop tests.

- a. Follow tests of normal function to see if the unit still has normal function.
- b. Document tests of normal function. Note if any functions have changed. Use the tests of normal function checklist template.
- c. For long-term tracking of units, record the life cycle history of the unit.

5. Execute drops on the closed and unlocked configuration.

- a. Set up the VxMark system in a normal closed and unlocked configuration, with the case latches unlatched (pictured below).



- b. Follow the standard procedure for drop tests on the unit (listed above in Step 2), except only on the bottom face since the case is unlocked. Inspect the unit for any changes, and document the results.

6. Execute drops on the open operational configuration.

- a. Set up the VxMark system in a normal open and operational configuration, with the screen pulled out (pictured below).



- b. Follow the standard procedure for drop tests on the unit (listed above in Step 2), except only on the bottom face since the case is unlocked. Inspect the unit for any changes, and document the results.

7. Check the unit for normal function after initial drop tests.

- a. Follow tests of normal function to see if the unit still has normal function.
- b. Document tests of normal function. Note if any functions have changed. Use the tests of normal function checklist template.
- c. For long-term tracking of units, record the life cycle history of the unit.

Results

1. Normal function was confirmed before drop tests.

- a. The structure of the unit before drop testing had no apparent issues. Photos of the unit before drop testing are attached in [Appendix A](#).
- b. Normal function was confirmed before drop testing. All essential functions with VxMark connected to the printer showed regular behavior and no problems, documented in spreadsheet: [Appendix B: VxMark Normal Function Checklist - 20260409 - Before drop test closed](#).
- c. Long-term tracking of the unit was logged.

2. The unit underwent standard bench handling drop tests.

a. Results for the standard closed configuration as listed below:

- i. After dropping on the bottom face (upright) along the 4 edges (4 drops):
 - 1. No damage was detected on visible parts.
 - 2. The headphones compartment door had opened, resting against the rear of the touchscreen (see photo below).



When lifting up the monitor, the door could spring open. The door could be closed again without issue. No cosmetic damage seen behind the touchscreen.

- ii. After dropping on top face (upside-down) along the 4 edges (4 drops):
 - 1. No damage was detected. No other issues found.
- iii. After dropping on the rear face (handle up), along the 4 edges (4 drops):
 - 1. No damage was detected. No other issues found.
- iv. After dropping on the front face (handle down), along the 4 edges (4 drops):
 - 1. No damage was detected. No other issues found.
- v. After dropping on the left face, along the 4 edges (4 drops):
 - 1. No damage was detected. No other issues found.
- vi. After dropping on the right face, along the 4 edges (4 drops):
 - 1. No damage was detected on visible parts.
 - 2. The headphones compartment door had opened again, resting against the rear of the touchscreen (see photo below).



The open headphones compartment door could be locked again without issue.

3. Normal function of the unit was confirmed after drop tests on the closed and locked configuration.

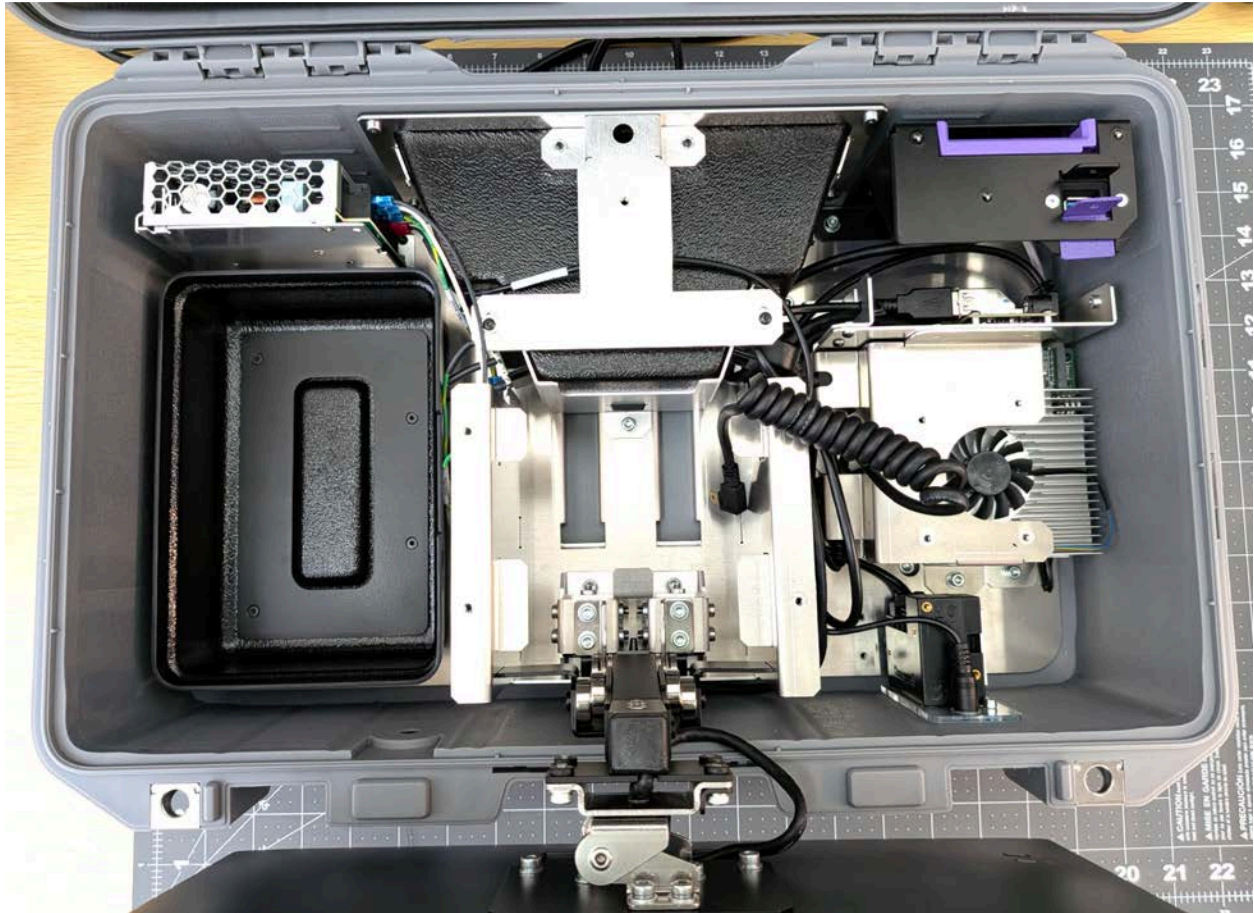
- a. The unit was reconnected to the test printer and started up.
- b. Normal function was confirmed after drop tests, documented in spreadsheet: [Appendix C: VxMark Normal Function Checklist - 20260410 - After drop test closed.](#)

4. Removable subsystems passed inspection.

- a. The headphones passed bench handling drops. They previously passed internal tests dropping from table to floor without damage. This is a more damaging drop than the bench handling drops.
- b. The printer was not tested, because it has previously been certified for IEC 62368-1 testing, which includes mechanical inspection and testing.
- c. The UPS was not tested, because it has previously been certified for UN 38.3 testing for transport of lithium batteries, which includes higher impacts than the bench handling drops in this procedure.

5. Observations internal to the unit after testing

- a. The unit was opened after testing to inspect fasteners, cables, and other internal components. No loose fasteners, cables, or other damage were found. No audible signs of loose parts were observed. A photo of the bottom tub is shown below:



8. The unit underwent additional drops in the unlocked and open configurations.
- a. There were no new findings. Drops caused no loss in function.
 - b. One instance occurred where the headphones compartment door opened after a drop while the case was open (pictured below):



The door could be closed like normal.

9. Normal function was confirmed after drop tests on the unlocked and open configurations.

- a. Normal function was confirmed before and after drop tests, documented in spreadsheets:
 - i. [Appendix D: VxMark Normal Function Checklist - 20260430 - Before drop test open](#)
 - ii. [Appendix E: VxMark Normal Function Checklist - 20260430 - After drop test open](#).

Conclusions

Effects of Bench Handling Drops on VxMark

- Bench handling drops are unlikely to cause functional failures of the VxMark unit.
- Bench handling drops are unlikely to cause any loosening of fasteners, cables, or other parts internal to the VxMark unit.

- Bench handling drops on a clean work surface are unlikely to cause cosmetic damage to the VxMark unit.
- The only noted effect was that the headphones compartment door opened 3 times out of the 8 groups of 4 drops each. They could be closed each time without issue.

Implications for VxMark Design and Implementation

Recommended followups are to:

- Instruct customers and poll workers receiving and using the units to check the status of that door when setting VxMark up after transport, to hold if needed and prevent it from springing open.
- Consider design refinements to prevent opening of the headphones door from shocks or impacts like those in this test.

Appendix A: Device photos before drops



Top



Front



Rear



Left



Right



Bottom, upright



Case opened



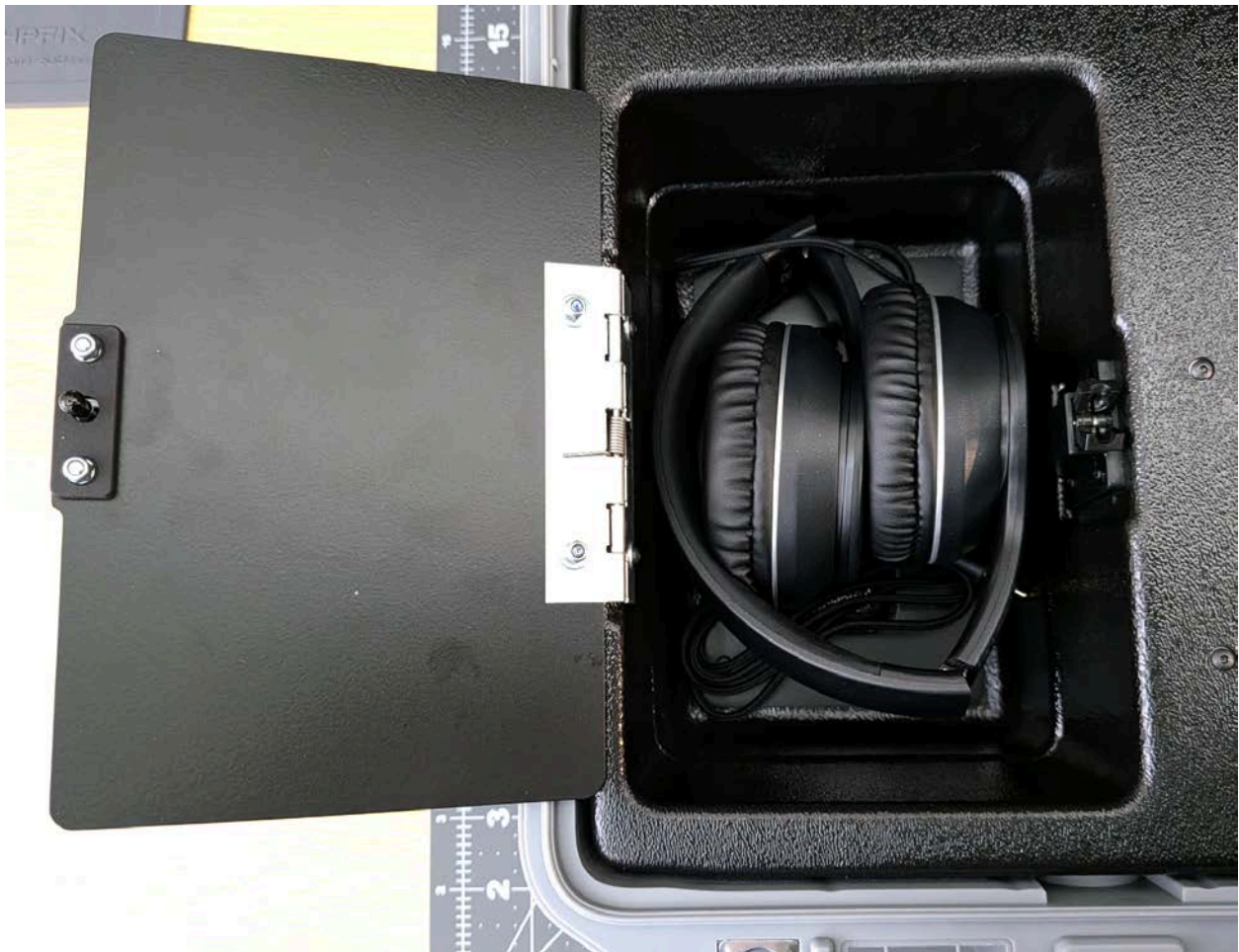
Operational configuration of touchscreen, front view



Bottom tub midplate below screen



Sticker seal below ATI was previously removed for thermal measurements and maintenance.



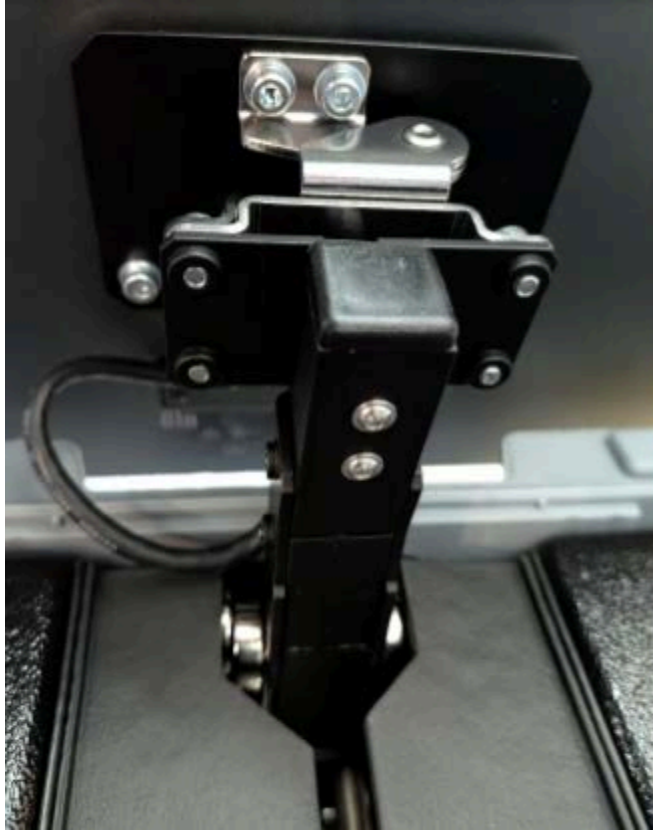
Left storage compartment for headphones with door opened



Rear cord storage compartment with door opened



Card reader and USB port access



Rear view of touchscreen support bracket in operational configuration

Appendix B: VxMark Normal Function Checklist - 20260409 - Before drop test closed

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius ...	VxAustin
	Date & time tested	4/9/2026	3:25 PM
	System tested & Unit/Model No.:	MK-14-018	
	Before what other test?	Drop test, bench handling	
	After what other test?	functional tests, card reader repair	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	Pass	No UPS since not part of test
2	Visually displays instructions and accurate feedback to users.	Pass	
3	Audibly plays instructions and accurate feedback to users.	Pass	
4	Accepts user inputs on the touchscreen.	Pass	
5	Accepts user inputs on the ATI.	Pass	
6	Accepts inputs on the barcode scanner.	Pass	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes
7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	Pass	

8	Protects internal electronics and subsystems.	Pass	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	Pass	opened up previously to examine card reader
2	Touchscreen	Pass	
3	Power Supply, internal	Pass	
4	Smart Card Reader	Pass	
5	USB-A receptacle	Pass	
6	USB 3 Switch	Pass	
7	Barcode Scanner	Pass	
8	Cable Assemblies	Pass	
9	Uninterruptible Power Supply (external to unit)	n/a	not used with unit
10	Printer (external to unit)	Pass	printer #2
11	Outer case	Pass	
12	Inner panels around electronics	Pass	
13	USB cover	Pass	
14	Security tie points	Pass	
15	Fasteners, both external and internal	Pass	
16	Wires, ports, and electrical connections	Pass	
18	other....		

Appendix C: VxMark Normal Function Checklist - 20260410 - After drop test closed

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius ...	VxAustin
	Date & time tested	4/10/2026	3:00 PM
	System tested & Unit/Model No.:	MK-14-018	
	Before what other test?	Thermals in warm ambient	
	After what other test?	Drop test, bench handling	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	Pass	No UPS since not part of test
2	Visually displays instructions and accurate feedback to users.	Pass	
3	Audibly plays instructions and accurate feedback to users.	Pass	
4	Accepts user inputs on the touchscreen.	Pass	
5	Accepts user inputs on the ATI.	Pass	
6	Accepts inputs on the barcode scanner.	Pass	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes
7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	Pass	

8	Protects internal electronics and subsystems.	Pass	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	n/a	not opened up since no functional issues
2	Touchscreen	Pass	
3	Power Supply, internal	n/a	
4	Smart Card Reader	Pass	
5	USB-A receptacle	Pass	
6	USB 3 Switch	n/a	
7	Barcode Scanner	Pass	
8	Cable Assemblies	n/a	
9	Uninterruptible Power Supply (external to unit)	n/a	not used with unit
10	Printer (external to unit)	Pass	printer #2
11	Outer case	Pass	
12	Inner panels around electronics	Pass	
13	USB cover	Pass	
14	Security tie points	Pass	
15	Fasteners, both external and internal	Pass	
16	Wires, ports, and electrical connections	Pass	
18	other....		

Appendix D: VxMark Normal Function Checklist - 20260430 - Before drop test open

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius ...	VxAustin
	Date & time tested	4/30/2026	2:30 PM
	System tested & Unit/Model No.:	MK-14-018	
	Before what other test?	Drop tests, bench handling, unlocked & open configuration	
	After what other test?	2-hour backup test, & storage tests	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	PASS	
2	Visually displays instructions and accurate feedback to users.	PASS	
3	Audibly plays instructions and accurate feedback to users.	PASS	
4	Accepts user inputs on the touchscreen.	PASS	
5	Accepts user inputs on the ATI.	PASS	
6	Accepts inputs on the barcode scanner.	PASS	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes
7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	PASS	

8	Protects internal electronics and subsystems.	PASS	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	PASS	
2	Touchscreen	PASS	
3	Power Supply, internal	n/a	not opened up to inspect
4	Smart Card Reader	PASS	
5	USB-A receptacle	PASS	
6	USB 3 Switch	n/a	not opened up to inspect
7	Barcode Scanner	PASS	
8	Cable Assemblies	n/a	not opened up to inspect
9	Uninterruptible Power Supply (external to unit)	PASS	
10	Printer (external to unit)	PASS	
11	Outer case	PASS	
12	Inner panels around electronics	n/a	not opened up to inspect
13	USB cover	PASS	
14	Security tie points	PASS	
15	Fasteners, both external and internal	PASS	not opened up to inspect
16	Wires, ports, and electrical connections	PASS	not opened up to inspect
18	other....		

Appendix E: VxMark Normal Function Checklist - 20260430 - After drop test open

Corresponds to detailed doc here:		VxMark Tests of Normal Function	
	Person(s) testing & location:	Pius ...	VxAustin
	Date & time tested	4/30/2026	3:30 PM
	System tested & Unit/Model No.:	MK-14-018	
	Before what other test?	Vibration tests	
	After what other test?	Drop tests, bench handling, unlocked & open configuration	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	PASS	
2	Visually displays instructions and accurate feedback to users.	PASS	
3	Audibly plays instructions and accurate feedback to users.	PASS	
4	Accepts user inputs on the touchscreen.	PASS	
5	Accepts user inputs on the ATI.	PASS	
6	Accepts inputs on the barcode scanner.	PASS	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes
7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	PASS	

8	Protects internal electronics and subsystems.	PASS	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	PASS	
2	Touchscreen	PASS	
3	Power Supply, internal	n/a	not opened up to inspect
4	Smart Card Reader	PASS	
5	USB-A receptacle	PASS	
6	USB 3 Switch	n/a	not opened up to inspect
7	Barcode Scanner	PASS	
8	Cable Assemblies	n/a	not opened up to inspect
9	Uninterruptible Power Supply (external to unit)	PASS	
10	Printer (external to unit)	PASS	
11	Outer case	PASS	
12	Inner panels around electronics	n/a	not opened up to inspect
13	USB cover	PASS	
14	Security tie points	PASS	
15	Fasteners, both external and internal	PASS	not opened up to inspect
16	Wires, ports, and electrical connections	PASS	not opened up to inspect
18	other....	PASS	headphones compartment popped open during test, still functional and closeable like normal